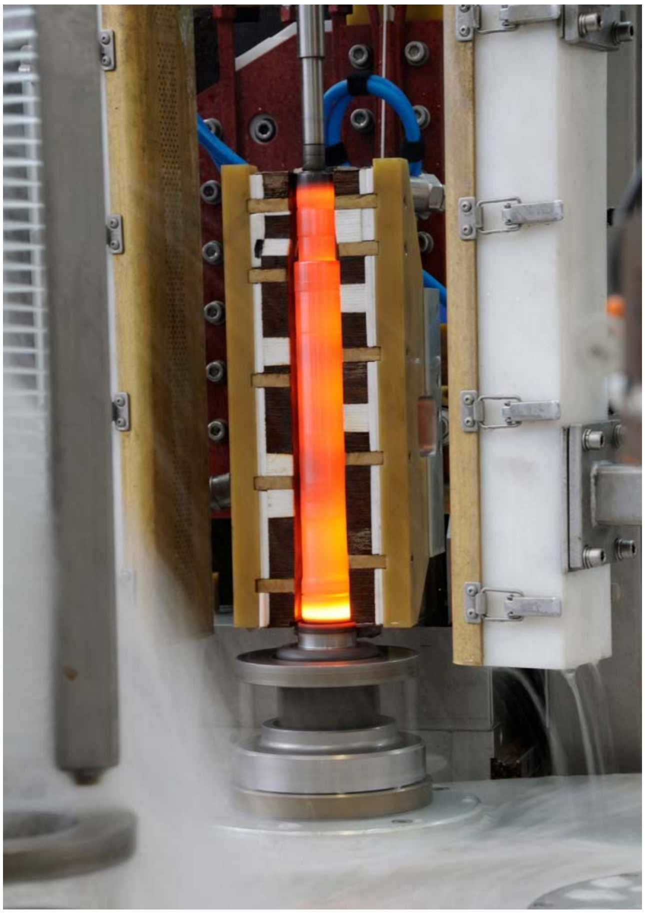
T/HSG INDUCTION HARDENING MACHINE



Customer:	HYUNDAI-WIA
Contact:	Mr. Taeho Lee
Inquiry No:	7002901.7
Workpiece:	Tripod Housing
Production Plant:	GH Electrotermia (Spain)
Date:	02/02/2017
Your contact in Sales Dept.	Pablo Arce
Phone:	+34 961352020 (ext 513)
Email:	parce@ghinduction.com

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1. Records

1.1. Introduction

7002901.7 version summarizes final commercial negotiation held on Changwon Hyundai WIA facilities on Feb 2nd.

7002901.6 is a last review including commercial conditions marked by Hyundai WIA for the project and adding an extraordinary price discount. Updated points have been marked in red for an easy and quick reading.

7002901.5 splits the prices according to Hyundai/WIA price list specifications chart.

7002901.4 is an update in the number of jigs and inductors according to the last RFQ received on 10th of October.

This review 7002901.3 includes the update in the number of jigs and coils as well as excluding a common supply cabinet from supply.

Updated modifications have been marked in green for an easy and quick reading.

Review 7002901.2 includes the following updates on the previous quote:

- Improved washing machine to a compact one.
- Modified lay-out.
- Included machine tools for all references.
- Service offer included

Updated modifications on this review have been marked in orange for an easy and quick reading.

7002901.1 was the update following the modifications held on last RFQ received on 8th of August and after the meeting hold in Hyundai-WIA Changwon in July.

This update includes mainly the following modifications:

- No tempering process would be included on supply.
- Input and output buffer are on inverse sense than previous lay-out.
- New lay-out reduced to L 6000 x W 6000 x H 4500 mm.
- All inductors referred to last part list are included on supply.
- Noise isolation up to <80dB is included as an option.

According to these changes the new lay-out has been considered as an installation all over a common platform without upper level. Flow of parts is just a reference as it will be defined after kick-off meeting. Lay-out will be sent attached on DWG file to this quotation.

Updated modifications have been marked in blue text for an easy and quick reading.

Following the request for quotation received from Hyundai-WIA for the supply of an induction hardening and tempering machine for tripod housing parts with final destination Monterrey (Mexico), and following the specifications of the document dated on ~~July 7th~~ August 8th 2016 signed by Mr. Hyun-Tae Park, please find hereafter our worked proposal.

The complete installation includes all the assemblies and devices required in the RFQ to follow completely its specifications for the treatment of short stem, long hollowed stem and female T/Hsg parts.

<u>Comments:</u>

- ~~In order to fit the installation in the specified lay-out an upper platform has been considered to locate some components such us power supplies, oscillator and control cabinets, the electric cooling circuit or the common supply connection.~~
- ~~Parts over 300mm length and 120mm diameter won't be able to be treated on tempering tunnel and they will be by-passed by a front conveyor on this station. Due to the particularity of hollowed long parts we recommend its tempering process in a furnace for getting better homogeneity results.~~
- In order to warrant the accomplishment of deformation specifications on long stem hollowed parts we have chosen its hardening by single-shot process which would allow getting 25" CT for all references on chart. GH Induction gets the compromise in case to be awarded with the project to work on the achievement of the application for these parts in our laboratory platform.

GH Induction is proud to introduce HYUNDAI-WIA this proposal getting our best technical and service solution:

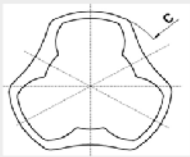
- Huge large experience on driveline heat treatment manufacturing with more than 300 references worldwide.
- High-tech exclusive solutions for induction and specifically driveline applications.
- Large experience in Mexico.
- GH Mexicana subsidiary company of GH Induction Group offers local technical service and quick intervention and spare parts supply.
- Fluent Spanish communication handled by our subsidiary in Mexico or our headquarters in Spain.

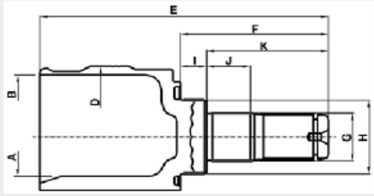
For all this and many more reason we have worked to make sure that GH Group offers to HYUNDAI-WIA the best and safest technical solutions. We remain fully available to describe, in detail, our offer and answer to any query you may have.

1.2. Project requirements

Part:	T/Hsg (Tripod housing)
List of parts:	

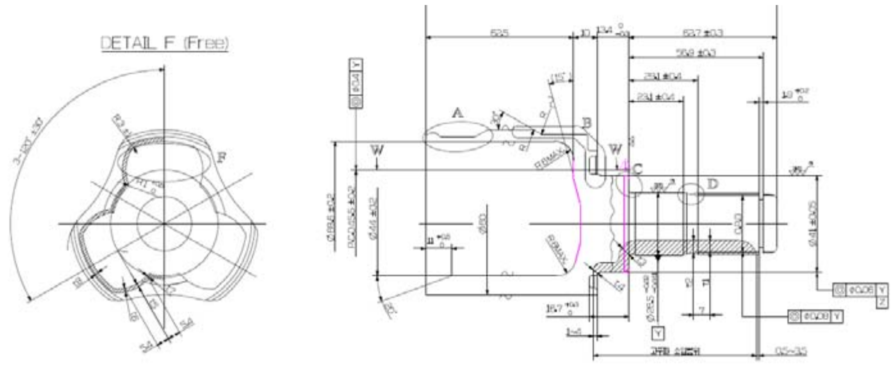
▣ T/HOUSING specification comparison



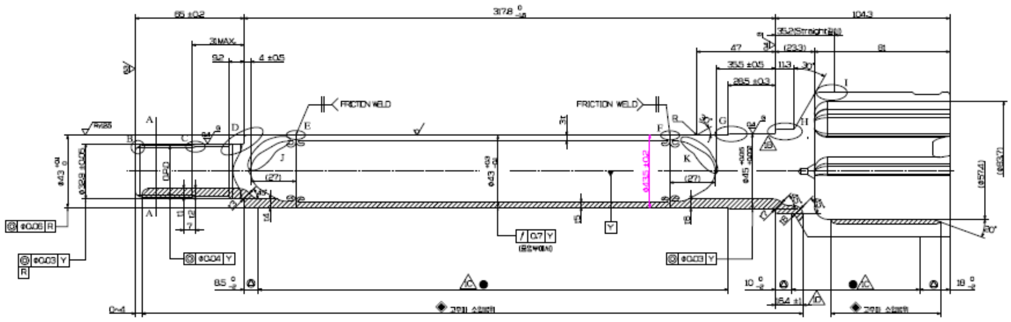


NO	Size	TYPE	Vehicle type	Item No	WORK Data												Remark
					A	B	C	D	E	F	Small G	Big H	Gap	I	J	K	
1	#19	TJ	D2UX	28019-D2001	Ø 88.8	Ø 57.8	Ø 66.7	Ø 66.7	162,1	59,1	Ø 27.28	Ø 84.95	7,75	21	23,1	57,6	
2			D2UX	28019-D2002	Ø 88.8	Ø 57.8	Ø 66.7	Ø 66.7	165,21	62,21	Ø 25.25	Ø 86.69	5,41	7,51	16,9	54,4	
3	#22	WTJ	HB	28022-HB601	Φ 46,0	Φ 70,0	Φ 77,5	Φ 77,5	160,1	76,1	Ø 26.5	Ø 41	14,5	12	28,1	62,7	
4		TJ	VDM	28022-016A	Φ 44,0	Φ 69,6	Φ 76,5	Φ 79,5	149,6	76,1	Ø 26.5	Ø 41	14.5	12	28,1	62,7	
5				28022-020	Φ 44,0	Φ 69,6	Φ 76,5	Φ 79,5	145,2	67	Ø 29.2	Ø 80.0	0,5	7,1	25,2	59,9	
6		CTJ	BDM	28022-P2701	Φ 46,0	Φ 65,6	Φ 72,5	Φ 72,5	146,7	67	Φ 29,2	Φ 80,0	0,5	7,1	25,2	59,9	
7	#23	CTJ	BDM	28023-P2702	Φ 47,6	Φ 67,4	Φ 74,9	Φ 74,9	152	62,1	Φ 23,5	Φ 41,0	12,5	18,4	28,1	56,9	
8		CTJ	BDM	28023-C9701	Φ 47,6	Φ 67,4	Φ 74,9	Φ 74,9	152,6	75,5	Φ 26,5	Φ 41,0	14,5	12	28,1	56,9	
6	#25	CWTJ	CX488	28125-C4901	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	181,4	84,5	-	Ø 60.7	-	-	-	45,4	FEMALE
7				28025-C4901	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	240,5	158	Ø 85	Ø 88	8	84	28	119	
8				28025-C4902	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	491	408,5	Ø 27.28	Ø 88	10,77	84,2	27,1	55,6	IDS
9				28025-C4903	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	240	152,5	Ø 28.2	Ø 85	5,5	28,9	26,6	98	
10				28025-C4911	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	197,5	110	Ø 29.92	Ø 88	5,71	29	85,2	76	
11				28025-C4912	Ø 52.6	Ø 76	Ø 84.5	Ø 84.5	471,5	887,6	Ø 29.92	Ø 88	5,71	25	54,7	57	IDS

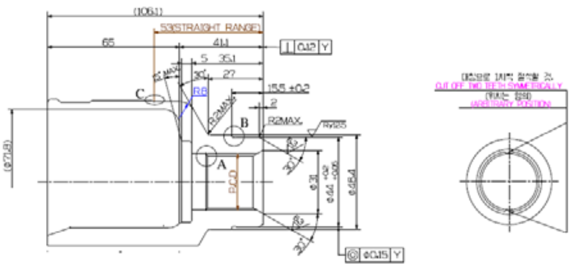
Main lengths chart



Short stem hardening pattern example



Long stem welded tubular part hardening pattern example



Female part hardening pattern example

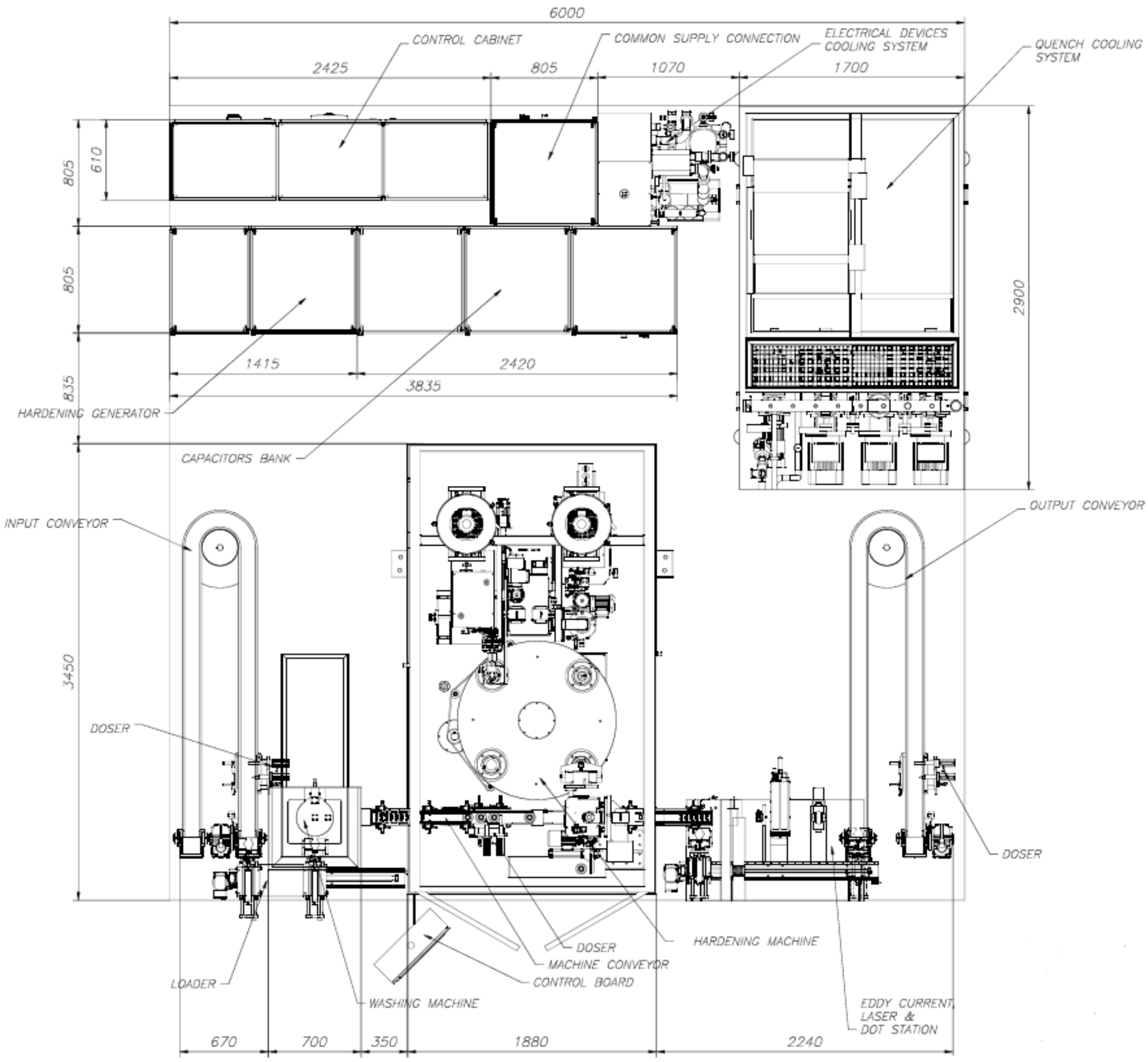
Material:	XC55, XC48, S55CKN, Cf53
Required cycle time:	25 sec for short stem and female parts // 40 sec for long stem parts
Machine capability:	Max Ø 130 mm – Max Length 500 mm
Surface hardness:	59~64 HRc after hardening~~/ 58~64 HRc after tempering~~
Microstructure:	Bell: all martensite / Stem: Perlite + Ferrite
Deformations:	Serration O.P.D. max 0.02 mm
Lay-out:	Within L6000mm x W6000mm x H 4500mm
Remarks:	The validity of the present proposal is not extensive to other parts or applications not specified here.

1.3. Proposal

The whole installation includes the following circuit:

1. Input conveyor with buffer for up to 30 parts.
2. Washing machine.
3. Hardening machine, indexing table type with 4 stations based in the following scheme:
    1) Loading/Unloading.
    2A) Positioning.
    2B) Bell's hardening.
    3) Stem hardening.
    4) Auxiliary quench.
4. Eddy current and marking station (laser and dot).
5. Output conveyor with buffer for up to 30 parts.

For the hardening machine has been considered a common 400kW power supply for bell and stems (in switched stations).



Lay-out

1.4. Production conditions

Availability

Cycle time ....25” for short stem and female parts / 40” for long stem parts
Production rate 95%

Installation site

Place, address..Monterrey (México)
Temperature, min. / max, ... 10-40ºC
Relative humidity, max, ...≤70%

Required supplies

Electricity ....3 x 480 V - 60 Hz
Industrial water ....20 ÷ 28ºC; 3 ÷ 7 bar
Compressed air ....5 ÷ 6 bar

REMARKS:

- If the customer has not proper water utility circuit within his plant to fulfil required features, GH could provide consulting services, on the basis of your instructions, and propose alternative solutions.
- If the max inlet temperature is above 28ºC, then a chiller unit with separate circuit for the converter is needed.

2. Scope of supply

The supplies related to the present proposal are:

2.1. Engineering

Induction application parameters study and test	YES	
Equipment and project engineering	YES	

2.2. Manufacturing

Components purchasing and equipment assembly	YES	
Assembly and put in operation	YES	
Technical documents	YES	
Pre-acceptance tests after manufacturing as itemized in § 8	YES	
Process statistic analysis & CPk[1] verification before delivery		NO

2.3. Delivery

Dismantling for transport and packaging	YES	
Transport and its assurances to final destination		NO
Handling in destination and placing in customer's factory		NO
Link to the customer's mains (AC supply, water, pneumatics…)		NO
Reassembly and put into production conditions on customer's plant	YES	
Training of customer's personnel	YES	
Final acceptance tests on customer's site	YES	
Technical support to the exploitation starting		OPTION

2.4. Standards

Standard GH: Installations Manufacturing

2.5. Technical documentation

As itemized in Standard GH: Installations Manufacturing	
Documents	5 manuals (3 copies in Spanish + 2 copies in English)

1. CPk: capability index intended to study the process repeatability and stability under normal work conditions

3. Equipment description

3.1. Input buffer and washing station

3.1.1. Input buffer.

The conveyors are made by aluminum profiles and stainless steel hinges, below hinges is placed a water collection tray of stainless steel pouring water into the washing machine. It has legs in aluminum profiles to support on the floor. The conveyor drive is performed by a motor reducer.

Input conveyor has been enlarged forming a U shape in order dispose a buffer capacity till 30 parts prior to access to the washing machine.

Fig.1 Diagram sketch from input conveyor to washing machine.

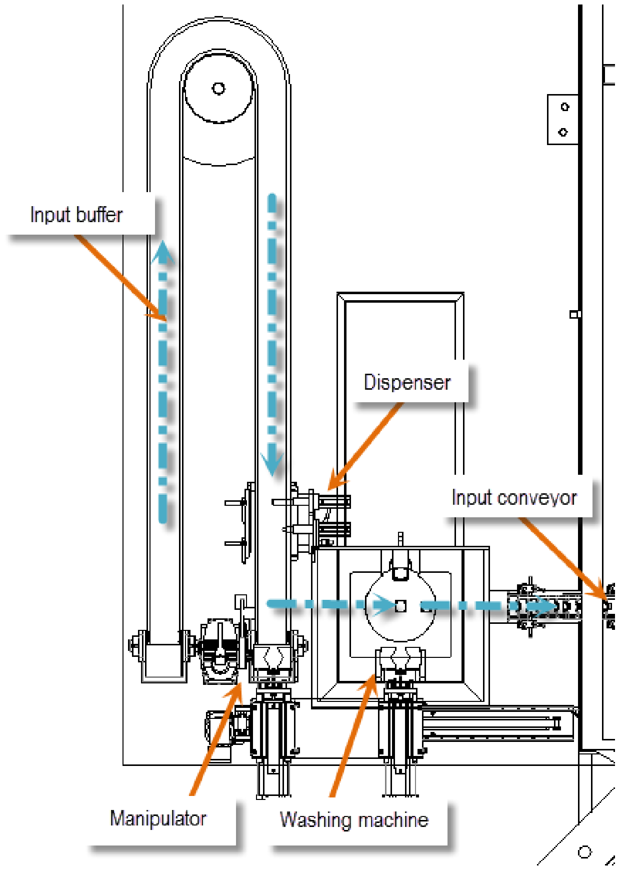
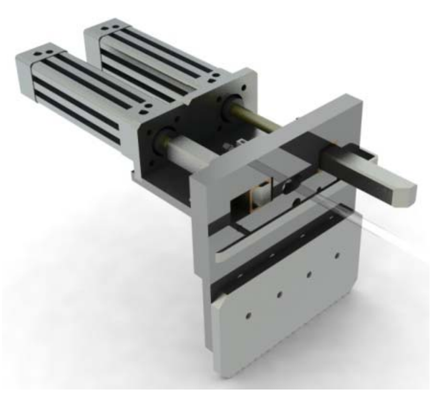


Fig.2 Conveyor.

Fig.3 Pneumatic dispenser.



Prior to the access to the washing machine it is mounted over the conveyor a pneumatic dispenser for let the parts pass one by one and so determine also the rhythm of dispensation.

A manipulator with pneumatic clamps loads the parts from conveyor to the washing machine in a horizontal movement.

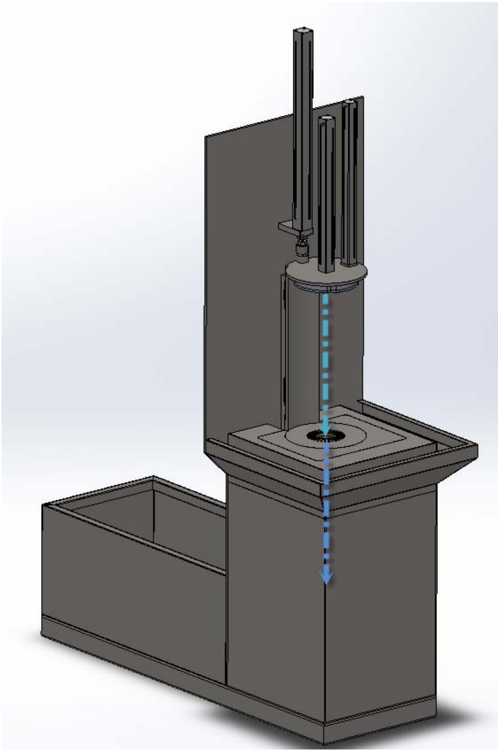
3.1.2. Washing machine.

Once the part is let over the washing platform, it is automatically took down to an internal tank where different showers, directly inside the bowl and outside, wash the part with hot water (30~60ºC) under pressure.

After the washing, an air blowing ring dries the part from top to bottom before lifting it again to initial position.

The washing machine includes a stainless steel tank of 500L with heat resistances, double bag filter and oil skimmer.

Once the cycle is done the manipulator takes the part to let it over the conveyor of the hardening machine.



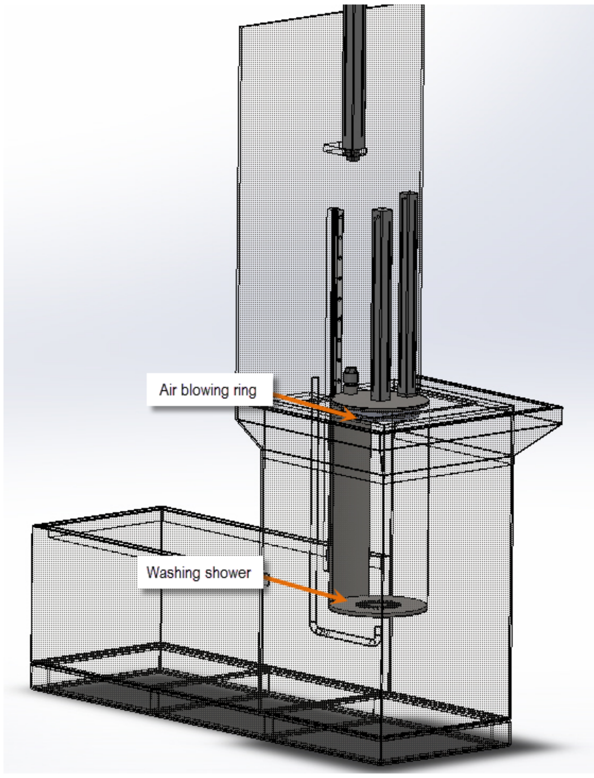
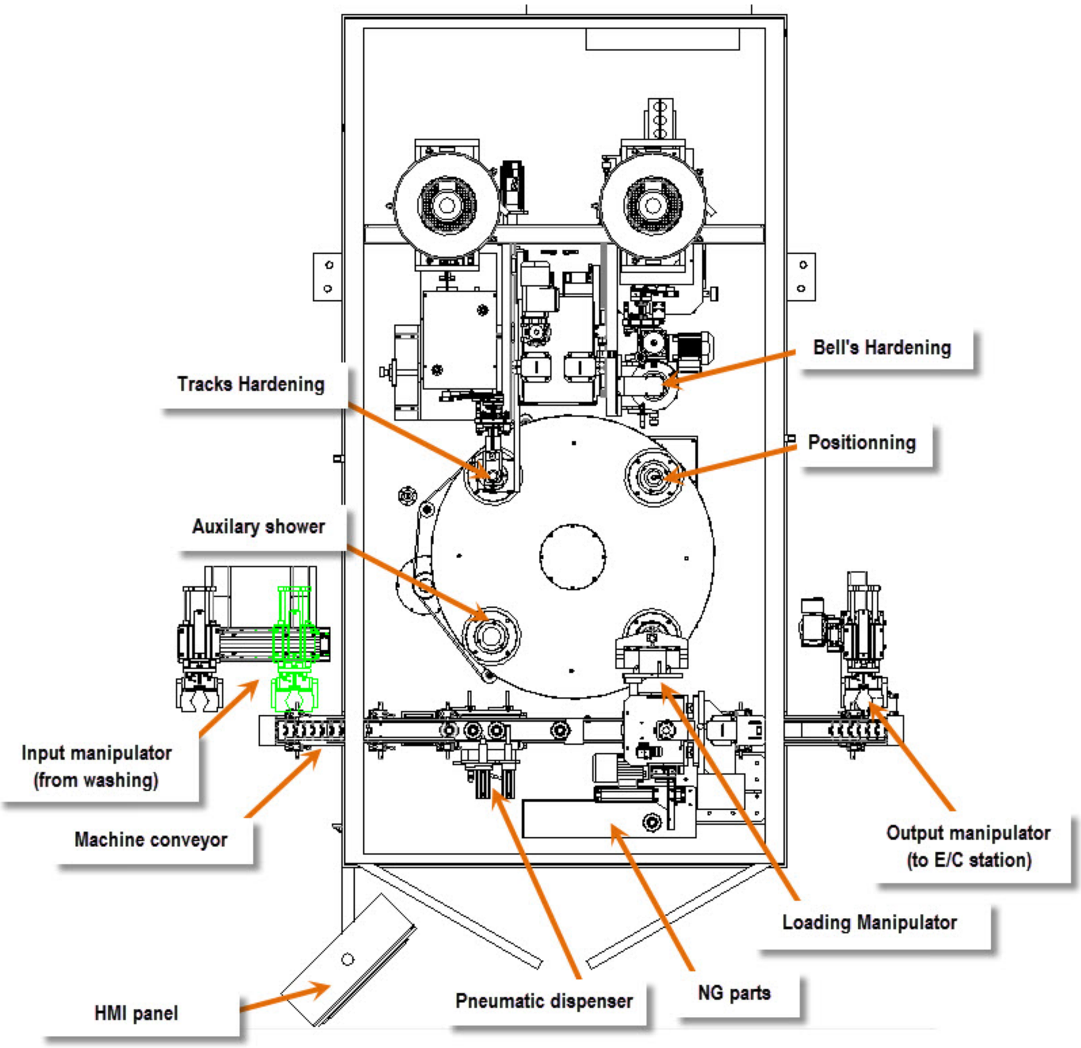


Fig.4 Washing machine movement sketch

3.2. Hardening Machine



3.2.1. Loading/Unloading manipulator

Prior to be taken by the manipulator a pneumatic dispenser spaces the parts at the entrance of the machine.

The loading and unloading manipulator is a turning arm with its vertical movement controlled by a servomotor and pneumatically driven on forward and backward movements. It is equipped with one gripper composed by two parallel self-centering strokes with analogical detection of diameters of parts.

The device is designed to be able to manipulate different diameters without any manual adjustment.

This manipulator transfers the parts from conveyor to Pos. 1 of the indexing table as well as brings it back once the part has finished the cycle.

In case that the part would be detected as NOK, the manipulator will reject it to a specific tray in the front of the machine.

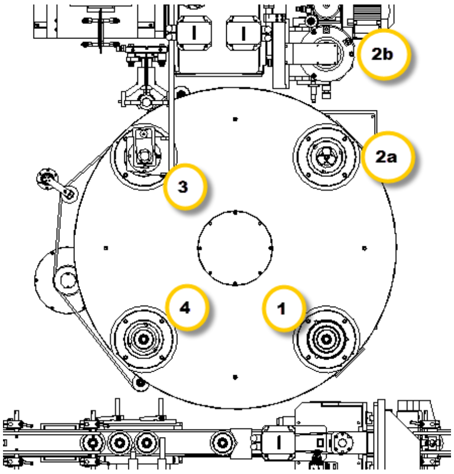
Fig.5 Loading manipulator self-centering clamps.

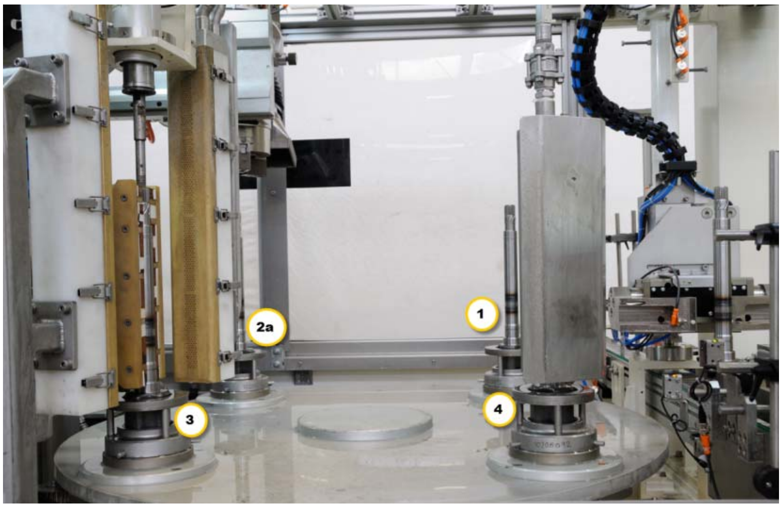
3.2.2. Hardening machine

The transfer of parts is carried out by a 4 positions indexing table. Every position has its corresponding tool composed by one fix and_another exchangeable part easy to replace for every reference. It is included the tools for one part reference in the scope of supply. Parts are handled with the bells downwards at any moment.

The table is responsible of the movement of the parts throw the different stations of the machine during the process by giving a part rotation when necessary performed by a gear motor and a system of pulleys and toothed belt.

Fig.6 indexing table diagram.





1. Loading/Unloading
2a. Positionning
2b. Bells hardening
3. Stem hardening
4. Auxiliary quench

When the part arrives to station 2a a jaw chuck picks it up and let it flow over the positioner, a servodriven device which positions the bell with high accuracy prior to its hardening. This jaw chuck is hanged to a column with vertical servo controlled displacement.

Fig.7 Jaw chuck and positioning device.

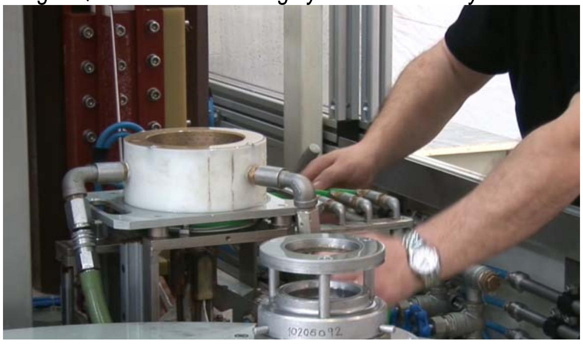
Once positioned the jaw chuck tights and fix its position and brings the part to station 2b (out of table) where the bell is hardened by single-shot without rotation in downwards position.

Once hardened, the jaw-chuck lets the part again over station 2a to let it go to next station and receive the new part.

The transformer-coil set is placed over a motorized XY table.

For an operator friendly change of reference the coil and quench set includes a rising system which simplifies its access and speed up the operation with safety.

Fig.8 Quench and coil rising system for an easy access.



On station number 3 a pneumatic tailstock fixes the part (its regulation is carried automatically) and the part is rotated by the tool. The tailstock includes a part rotation sensor and a height check device to check the reference and deviations on height before and after hardening.

The inductor and quench set is approached to the part to carry on the hardening of the stem. Even if the stem hardening for this proposal has been considered by single-shot for short and long stem parts, the machine is prepared to do the process whether by single-shot or scanning.

Fig.9 Stem hardening station.

The transformer-coil set is placed over a motorized XY table.

The hardening of female parts it's also carried on this station. For this purpose it has been included an auxiliary transformer set to give a high frequency rate (70~100 kHz) for a proper hardening of this area. This hardening can be done even by scanning or single-shot process.

On station number 4 the part receives an auxiliary quench to finalize the process and cool down the part prior to its unloading.

Fig.10 Auxiliary quench.

The part backs to station number 1 where it is carried out by the manipulator again to the conveyor.

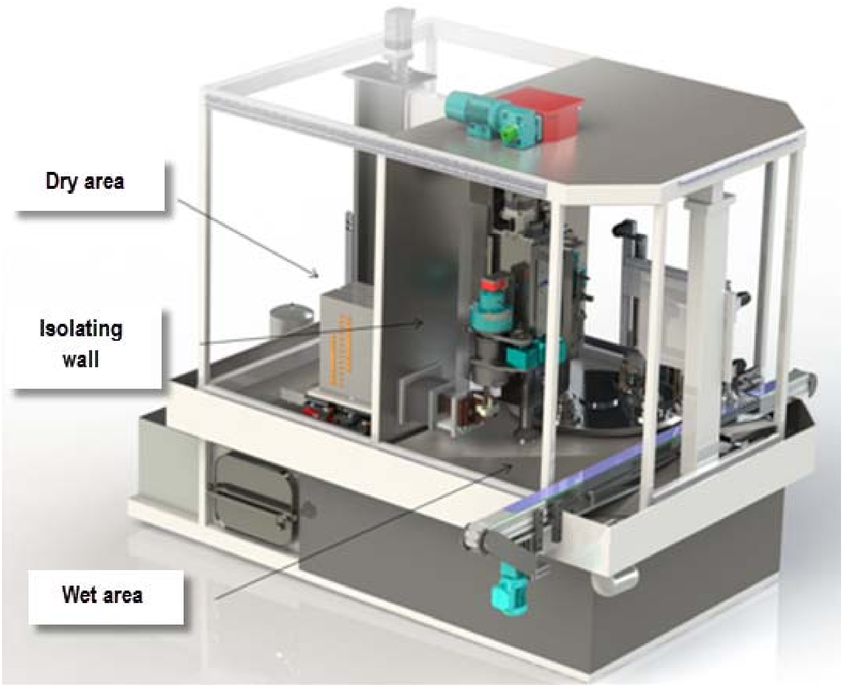
Over the output conveyor it is installed a drying curtain by compressed air to rinse the parts prior to come to the Eddy Current station.

Note: This auxiliary quench and blowing curtain has been confirmed by the Eddy Current manufacturer to be adequate for a proper measuring on its device so no additional washing or drying operation should be required.

To avoid corrosion and degradation on elements in contact with the quenching water, the machine has been designed with an isolating wall which separates a “Dry” and “Wet” areas.

So splashes and vapors contact is minimized with critical components such as transformers, XY tables or ball screws placed in the dry zone which clearly enlarge their life.

Fig.11 Dry and Wet area isolation



3.3. Eddy Current - Marking station and Output buffer

Parts are taken from hardening conveyor by pneumatic manipulator which hangs them through the whole station.

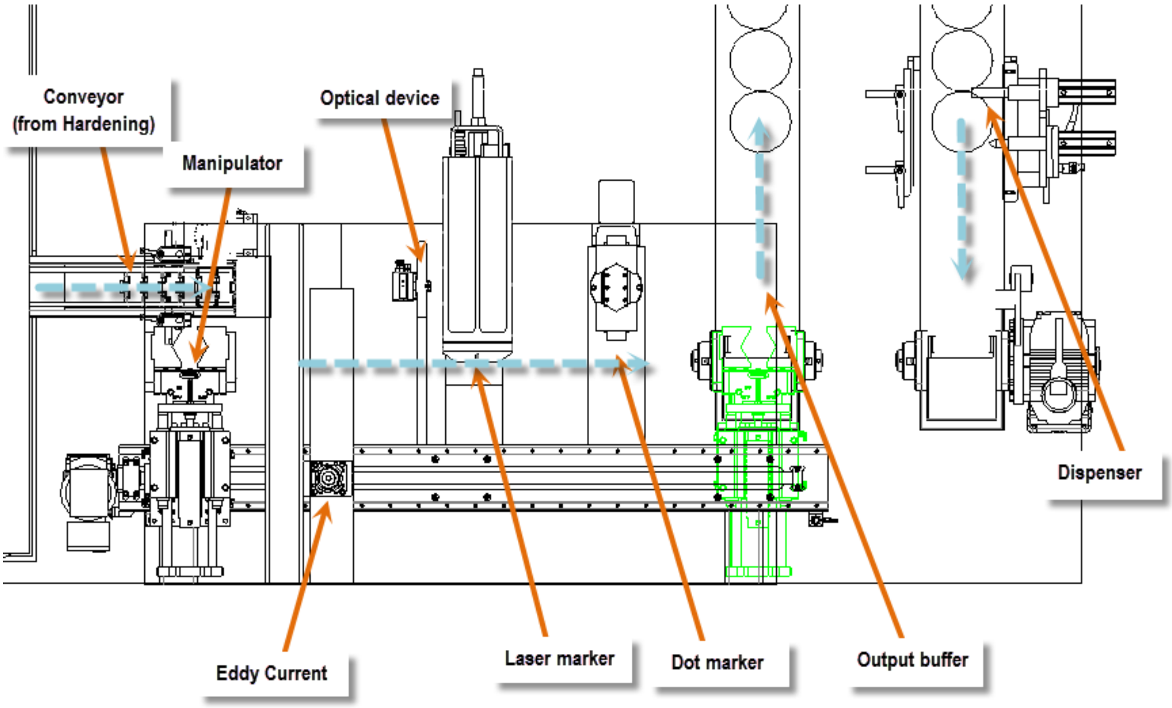
Eddy Current brand IBG model Eddy-Visor DS 2 channel checks bells and stem hardening and material. All sets of masters described on RFQ has been included.

Laser marker brand Keyence model MD-X1000 marks the part at the described position on the head. The correct position of the part is handled by an optic camera sensor.

Dot marker JEIL MTECH Marksman MCU-100N Model marks the part at the described position on the bell. The correct position of the part is handled by an optic camera sensor.

Once checked and marked the parts are left over the output conveyor which has been enlarged forming a U shape in order dispose a buffer capacity till 30 parts prior to access to the next machine process. A pneumatic dispenser will let the pass the parts in a constant cadence.

Fig.12 Eddy current and marking station.



*1 *Note: Noise level close to main pumps and dot marking station won't accomplish <75dB noise level requirement without an acoustic capsule isolation.*

3.4. Electric system, control and command devices

1. Control system

The main function of the control & command system is the machine running and managing always subordinated to the safety of personnel and environment.

The hardening and tempering machines have independent control systems.

In the hardening machine is composed by:

Computer Numeric Control (CNC).................. Siemens 840 Dsl, PCU 50, with 9 axles servo controlled (3 of them controlled by CNC)

Man - Machine Interface (MMI) .... Siemens OP12

Hand held - Machine Interface (HMI) .... Siemens HT2

2. Quality process supervision system

The operation concerned by this proposal is controlled by supervision of the following parameters:

MF/HF output frequency..	YES	
MF/HF output power..	YES	
Energy delivered during the heating...	YES	
Heating time	YES	
Product cooling water flow	YES	

Product cooling water temperature	YES	
Quenching time	YES	
Product rotation	YES	
Product position and/or speed..	YES	
Coil position and/or speed..	YES	
PMS - GH Process Graphic Control.......................................	YES	
Datalogger...	YES	

Parameters supervision:

- Integrated in control & command system (CNC o PLC)
- Performed by independent system

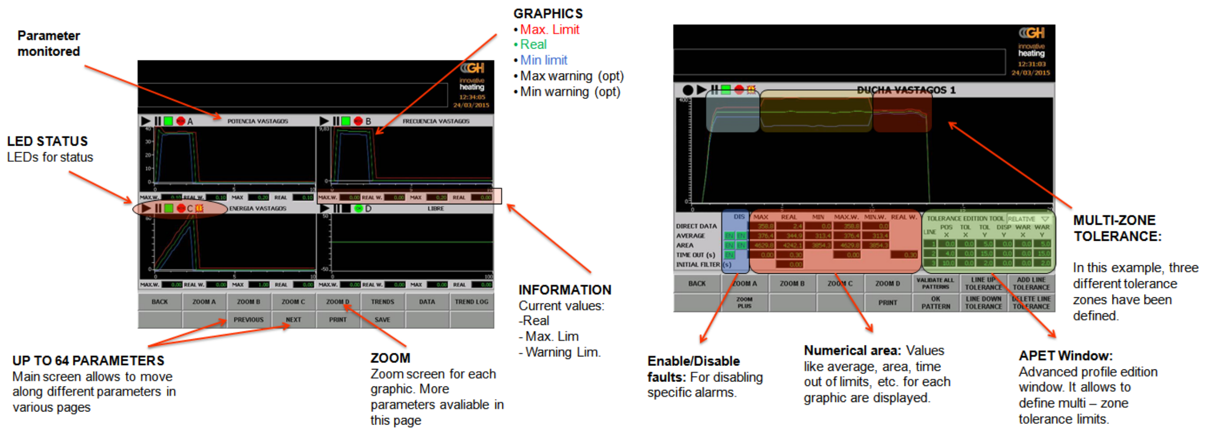
In case of non–conformity with the reference values of one or several parameters, the quality following system

Alerts the operator with an alarm signal	YES	
Starts the product automatic separation.................................	YES	
Stops the machine..	(User setting)	

Some screen examples of PMS - GH Process Graphic Control

GHPGC is the process monitoring system (PMS) developed by GH that allows to monitor and to control main heating process parameters in the safest and most functional way.

Fig.18 PMS screen examples.



It is carried out the graphic control of up to 99 parameters such as (*depending on the sensors and devices installed in machine):

Heating process	Cooling process	Hardening depth
Heating time	Quenching time	Frequency
Part temperature	Quenching flow	Heating time
Part energy	Quenching temperature	
Power	Quench polymer concentration	
Voltage		
Current		
Frequency		

In order to evaluate if a process is valid main parameters are recorded and plotted in real time according sampling time during whole cycle. When a good part is validated, MAX and MIN limits are calculated and plotted for each parameter in accordance to the tolerance values defined by the customer. After this moment, parts produced are monitored and compared with the defined limits. If any parameter detected is out of the limits, an alarm is triggered and part is considered as a NOK part.

3. Generalities

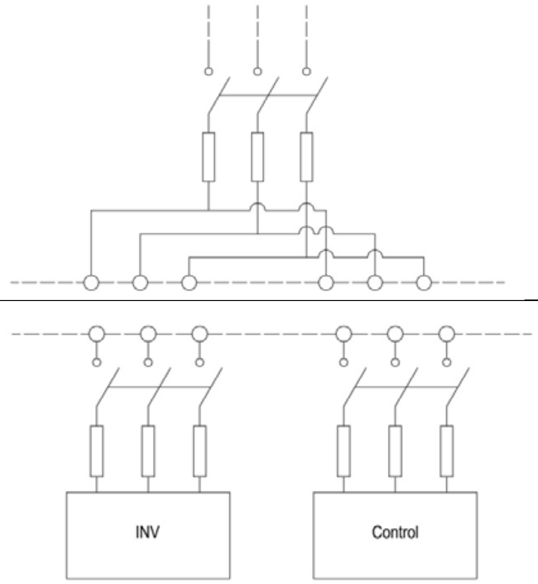
The electric system is made in accordance with standard GH NG 1.7

The execution of electric inlet includes two independent lines:

- One, for the induction equipment (generator and peripherals) and
- Other, for the general control system, mechanisms and electric devices in general.

This solution allow to keep a part working while the other is disconnected, functioning mode necessary for induction generator adjusting, maintenance and repairing operations.

This double inlet is compatible with an eventual common line, usually installed by the customer, with the usual protection, measure and/or breaking elements.



Common inlet, installed by

Customer

Works included in the present proposal

INV: Induction system.
Control: machine control & command board

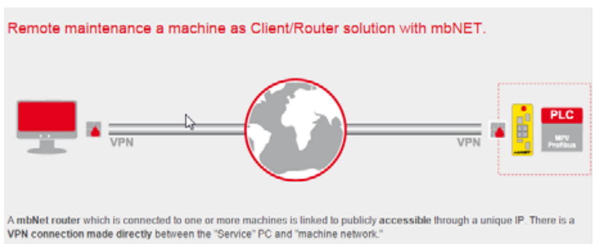
The protection grade of boards is:

IP	Material
54	Electric boards in general: generator, oscillating circuit, command panels

3.5. GH Remote assistance service

With the aim to assure a better service meanwhile a remote assistance service, GH installs a wireless router in the machine which allows a remote access to our technical service staff to solve possible incidences directly in the PLC from our own facilities in GH Electrotermia (Spain). This new tool would help us in a quicker answer or in case of any technical assistance.

The access to the machine is done throw a secure VPN connection by a dedicated hardware (WAN or 3G connection). Internet SIM card is not included on supply.



This service will be given free of charge during all the warranty period. After this time the customer may return the hardware to GH or buy it for 1.700,- €. The service fee for a 24h assistance during a year (after warranty period) will be of 1.000,- € and in case of technical assistance requirement the charge will be measured per hours (90€/hour).

3.6. Transistors induction generator (Series)

3.6.1. Functioning

The induction generator is the system destined to convert the energy coming from the industrial AC mains in a medium or high frequency signal adapted to the feeding of the induction heating station. This signal is a square wave that goes through an oscillating circuit (capacitors bank and an inductor as a minimum) and origins a magnetic field. Once that a part is under this magnetic field, electrical currents are made upon it. These currents increase the superficial temperature of the part by means of the Joule effect.

To obtain the square signal, the AC industrial current is rectified to a DC current. After this, a transistors inverter bridge obtains the desired AC wave at the adequate frequency for the application.

3.6.2. Security

The ensemble formed by generator and oscillating circuit is water refrigerated. The equipment protection is granted by the Reading of the integrated flow-meters. The control electronics keeps always protected the semiconductors against internal and external failures (inductor breakage, short-circuits, overloads…).

GH generators accomplish with Machine Security Standard ISO 13849-1, category PL d

3.6.3. Technical data

The selected generators are 400SM20 for the hardening, being its power 400 kW. The following chart shows the main features of SM generator series.

Type		12SM	25SM	50SM	75SM	100SM	150SM	200SM	250SM	300SM	400SM	500SM	600SM	800SM
Output continuous power	kW	12	25	50	75	100	150	200	250	300	400	500	600	800
Frequency	kHz	[0.5, 20] [20, 150]					[0.5, 20]							
Power supply	kVA	15	31,25	62,5	93,75	125	187,5	250	312,5	375	500	625	750	1000
Voltage supply		380–480 VAC 50/60 Hz			380-480 VAC 50/60 Hz (1)									
Generator width	in/mm	23,6/600			31,5/800		47,2/1200		55,1/1400	55,1/1400		102,3/2600		118,1/3000
Generator depth	in/mm	23,6-31,5/600-800			31,5/800		31,5/800		31,5/800	31,5/800		35,4/900		35,4/900
Generator height	in/mm	43,3/1100			78,7/2000		78,7/2000		78,7/2000	78,7/2000		78,7/2000		78,7/2000
Width with cooling system	in/mm	23,6/600			78,7/2000		78,7/2000		86,6/2200	90,5/2300		137,8/3500		157,5/4000
Depth with cooling system	in/mm	23,6/600		31,5/800	31,5/800		31,5/800		31,5/800	31,5/800		35,4/900		35,4/900
Height with cooling system	in/mm	74,5/1900			78,7/2000		78,7/2000		78,7/2000	78,7/2000		78,7/2000		78,7/2000
Water temperature min/max	ºF/ºC	[68, 98,6] / [20, 37]												
Water supply		1''								1'' ¼''		1'' ½''		
Waterflow	gpm/lpm	2,6/10			3,9/15		7,9/30	8,8/40	12,7/48	18,5/70		30,38/115		36,7/140

3.6.4. Generator interface

The induction generator allows its control and command interface through an 8'' panoramic colour touch screen (TouchHMI). Its user friendly interface makes able to supervise the equipment general functioning and to control the output signal.

Fig.18 Power supply screen examples:

A: Main screen

B: Flow-meters visualization

C: Process parameters configuration

D: Configuration screen

The interface allows the user to visualize the state of the equipment (prepared, heating, fail); visualize alarms (25 last events); configure the functioning parameters of the equipment. All this is done on an easy and intuitive way, "surfing" through different screens:

- Visualization of generator's analogical parameters: power, current, voltage, frequency and energy. Power regulation set.
- Screens related to the watch of flow-meters.
- Screens for parameters configuration and functioning mode.

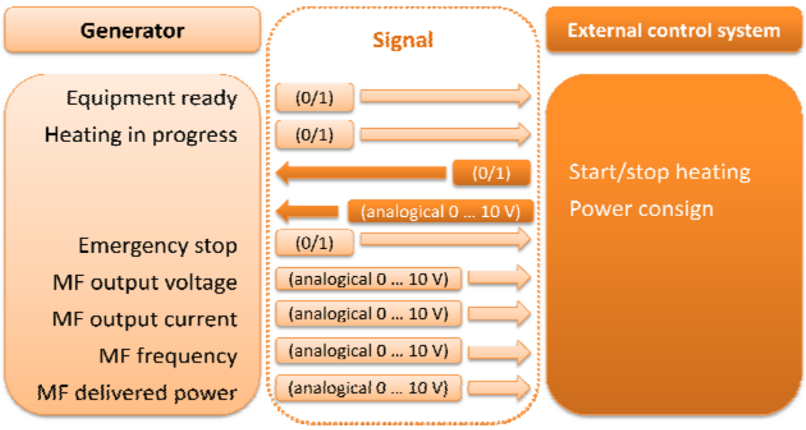
Optionally, not included on the base supply, the generator can store programs to implement temperature controlled automatic heating cycles.

Also optionally, not included on the base supply, the generator can work as the main control for a simple machine by means of an internal program system (IPS) developed by GH. This program must be done by GH personnel.

3.6.5. Connection to an external control system

The generator is also ready to be controlled from an external system, so that the induction heating equipment is integrated into a system controlled by a general device. In order to do this, it could be linked through bus connections for its external control by means of: Profibus, Profinet, Device-net, Interbus.

Optionally, the generator can be equipped with different analogical inputs / outputs destined to exchange the following signals:



3.7. Oscillating circuit

The capacitors bank is the element providing the necessary capacity to establish, with the self-inductance of the inductor eventually augmented by the impedance adaptation transformer when applied, the resonant output circuit which will define the frequency of the induction signal.

Essentially, the oscillating circuit consists of one or several capacitors chosen depending on the oscillating conditions to state.

The capacitors are built in mixed dielectric, biodegradable, free of any pollution danger. The permissible overload is 1,35 the nominal current, including harmonics.

The oscillating circuit includes a coil interface transformer. This is a device intended to increase the coil self-inductance, when this value becomes too low to compose an oscillating circuit with an appropriated level of power allowing the induction heating.

The working principle of this transformer is based on the multiplication of the coil self-inductance, L, by a value, n, given by the transformer ratio.

Depending on the required ratio, the working voltage, power and frequency to apply, the transformer may be manufactured under different versions but it is always made under an isolating and protection cover.

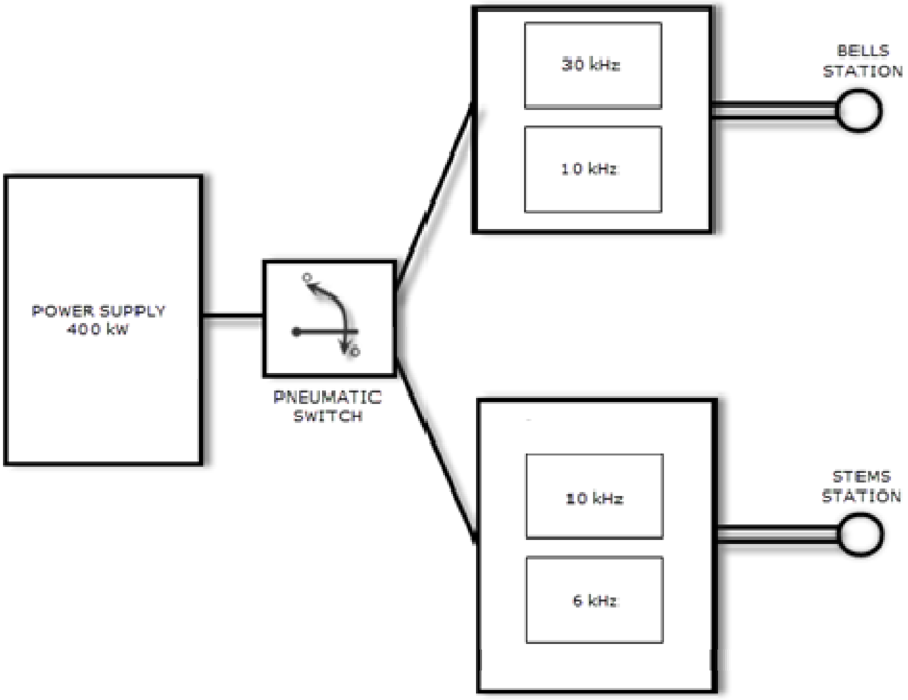
Constructive solution details

Capacitors bank with adjustable capacity.. YES
Oscillating circuit under IP54 cover.. YES
Oscillating circuit equipped with coil interface transformer.. YES
Hardening quench circuit equipped with quick connection plugs.. YES
Coil interface transformer equipped with adjusting taps.. YES

The safety against electric risks for access to zones under voltage is guaranteed by:

- Occasional access zones: panels screwed or fixed by other elements needing tools for dismantling.
- Doors: lock with key plus opening detection with automatic disconnection of the power

Fig.19 Oscillating circuit scheme



3.8. Coil or inductor

The coil or inductor is the element in the induction heating system performing the direct action on the product to process, it means: it is the component destined to transfer the energy to the material to treat. With this aim, the inductor is built with the form and size appropriated to the product.

The coil is manufactured as an assembly rigid and stable against the mechanical or thermal efforts, allowing a repetitive and controlled operation and a free of errors placing and positioning.

The main parts are:

Coil: with a turns number and shape specially chosen to obtain maximum efficiency. The coil is built with copper pipe intended to operate as internal cooling circuit destined to evacuate the residual heat due to the Joule effect loses.

Fastening system: it works as electrics connectors and as mechanical support and positioning.

Cooling circuit fitting: it is the water input / output elements connecting the coil to the coolant circuit.

The coil is equipped with a cooling quench destined to refrigerate the product after the action of the heating coil. This quench is built with magnetic inert materials in order to avoid any effect on inductor field.

<u>Constructive solution details</u>

Coil equipped with fast connecting system..	YES
Coolant circuit equipped with fast connecting system	YES
Coil equipped with cooling quench	YES
Cooling quench equipped with fast connecting system	YES
Quench included in the machine	YES

<u>Non-contractual examples of different coils to be applied on this machine:</u>

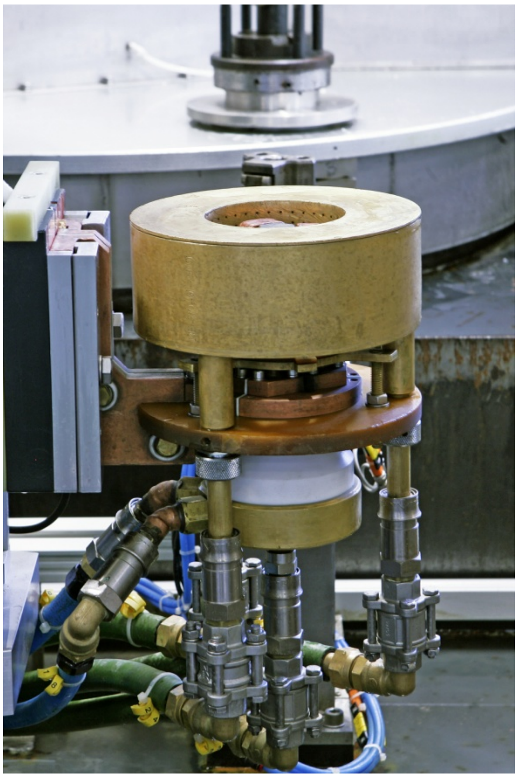


Fig. 20: Bells single-shot coil and quench

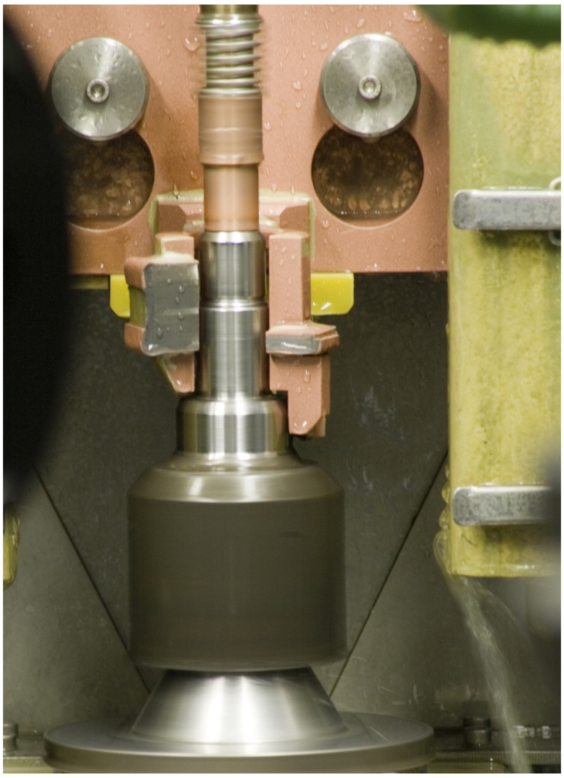


Fig. 21: Short stem single-shot coil and quench

Fig. 22: Long stem single-shot coil and quench

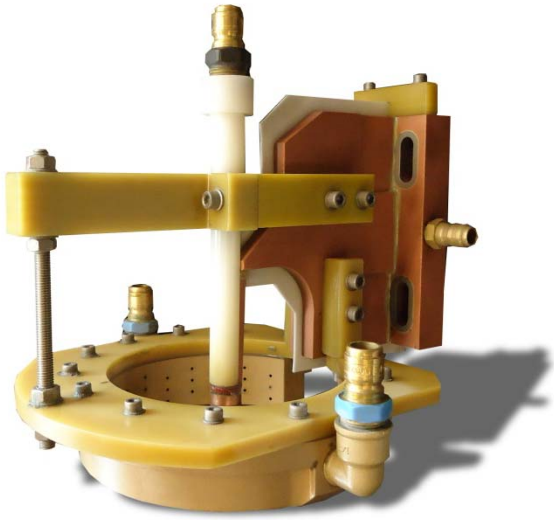


Fig. 23: Female scanning coil and quench set

Following the last drawings these are the previewed inductors to be used:

Bell (6 inductors)

Ind	Drawing Reference
#1	23019-D2001 / 23019-D2002
#2	23022-HB015A / 23022-020
#3	23022-HB501
#4	23125-C4912 / 23025-C4901 / 23025-C4902 / 23025-C4903 / 23012-C4911 / 23025-C4912
#5	23022-F2701 / 23022-F2702
#6	23023-C9701

Short Stem (8 inductors)

Ind	Drawing Reference
#1	23019-D2001
#2	23019-D2002
#3	23022-HB501
#4	23022-015A
#5	23022-020
#6	23022-F2701
#7	23022-F2702
#8	23023-C9701

Long Stem (5 inductors)

Ind	Drawing Reference
#1	23025-C4901
#2	23025-C4902
#3	23025-C4903
#4	23025-C4911
#5	23025-C4912

Female (1 inductor+quench set)

Ind	Drawing Reference
#1	23125-C4901

3.8.1. 3DPCoil

GH Group has patented a new system for inductors design and manufacturing using exclusive 3D printing technology. Results in comparison with the traditional manufacturing process are:

- Reduction of costs per part: 3DPCoils have a longer working life thanks to its solid design in one single part free of weldings, which revert to a highly increased productivity. Additionally to this, as their geometrical repetivity is guaranteed, its maintenance and recalibration time are minimized.
- Reduction of stocks: The increase in cycles of the inductors combined with the reduction in delivery time result in a reduction of spare stock.
- Savings in maintenance: The increase in cycles reduces the consumption of inductor spares. Furthermore the shorter reference changes minimize the labour time required for such changes.
- Quality of the parts: The ability to manufacture and supply identical coils ensures that heating profiles are maintained when fixtures are changed. This technology allows also complex coil's designs impossible to conceive by handcraft manufacturing.

GH 3DPCoil (patent) printed in one unique part.

GH Microfusion (patent) inductor with embedded quench.

GH based on the application will decide what inductor's technology may be used.

3.9. Induction heating cooling system

Some elements working in the induction heating system (e. g., transistors, capacitors, inductors) under the action of high currents require cooling to evacuate the heat by Joule effect.

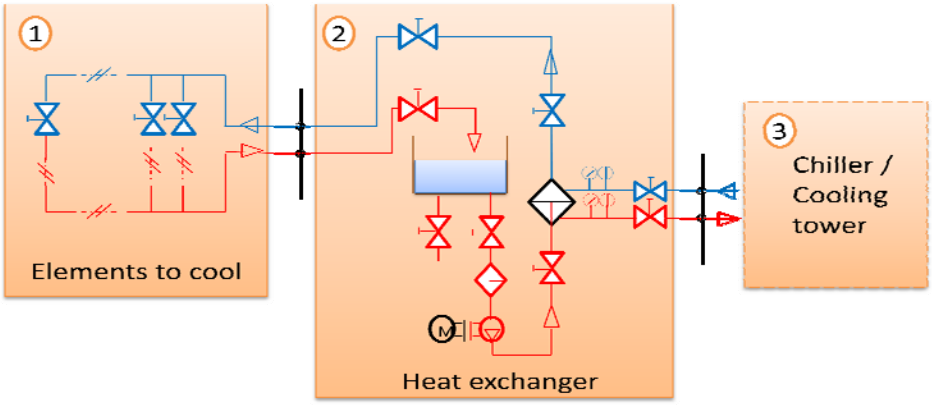
This cooling is based on a closed loop of liquid coolant through these elements. The system is named by GH as R3.

The liquid coolant applied in this circuit is demineralised[2] water. The low content of salts reduces the risk of obstruction by solids.

The coolant itself is cooled in a heat exchanger, where it transfers the heat to a second coolant, usually industrial water (water without any special chemical requirements).

Finally, the industrial water is cooled in a cooling tower or in a frigorific machine or chiller.

2. Avoid all confusion with distilled water which is not appropriated for this purpose because it acts as an acid due to its tendency to ionization under the temperature and voltage conditions present in the cooling circuits and in contact with the metallic parts of such circuits. The demineralized water is practically inert, dielectric and non-polluting (no risks are associated to the lose)



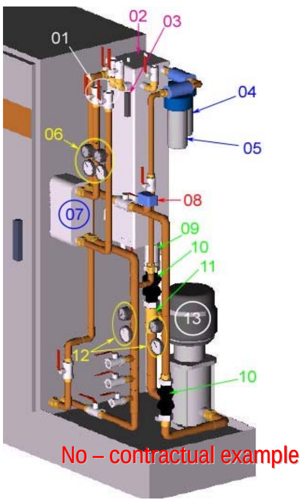
These three sections are independent systems:

1. The coolant circuit is integrated in the induction system and it is composed of a circuit with its appropriated control and measurement devices (valves, flow meters, thermostats…). Its function is to conduct the coolant through the components to cool in order to guarantee the induction system working temperature. In case of lack of capacity (due to coolant failure or excessive temperature), the integrated protections stop the equipment under failure condition.
2. The R3 unit is a system intended to maintain the coolant temperature as a thermal interface between the coolant circuit and the industrial water network
3. The external cooling equipment (usually a cooling tower or a chiller) is a part of the industrial water network not included in the induction heating equipment.

See in the chapter concerning the scope of supply the parts included in the present proposal.

3.9.1. R3 unit – demineralised water cooling system

The cooling R3 system is a well proven standard design of GH, fitted into most GH machines and generators. Treated water is used as coolant for the equipment in a closed circuit and is cooled at its turn by industrial water as secondary coolant in open circuit. The heat exchanger is of plate type.



01 : Industrial water input/output
02 : Upper access to the cooling water tank
03 : Level indicator
04 : Polypropylene filter
05 : Deionising resin vase
06 : Thermometer & manometer
07 : Heat exchanger
09 : Output valve
10 : Attenuating connectors
12 : Thermometer & manometer
13 : Pump

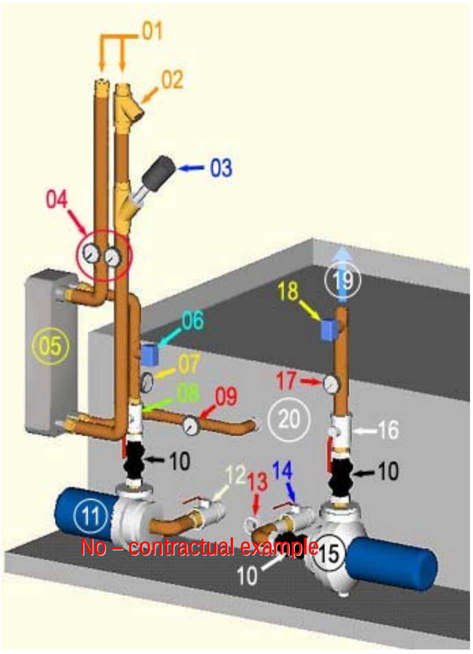
3.10. Hardening liquid cooling circuit

The product cooling water is applied on the product by quenching through a specific quench. The flow and temperature are measured in order to guarantee the process stability and repeatability.

A pump installed on the circuit origin keep the water circulation. After quenching, the water is recuperated by means of a recycling in a closed loop after conditioning. This system is named R4 by GH.

3.10.1. R4 unit - Description

The system is specially built to work with the usual product coolant as water containing soluble polymers additives. The excess of heat is taken out from the coolant by an industrial water flow through a heat exchanger. The standard construction and main components is:



01: Input / output industrial water connections
02: Y - type filter for protection of circuit against solid elements
03: Pneumatic switch valve (component applied eventually)
04: Manometer for primary circuit verification
05: Heat exchanger (remark: exchanger represented on the schema is standard welded type, depending of the specific conditions or under request, this exchanger may be dismountable type)
06: Product cooling water flow detector.
07: Product cooling water thermometer (exchanger inlet side).
08: Pump isolating valve (outlet side).
09: Product cooling water thermometer (exchanger outlet side).
10: Elastic connector for vibrations attenuation.
11: Product cooling water pump.
12: Pump isolating valve (inlet side).
13: Product cooling water tank outlet.
14: Quench isolating valve (inlet side).
15: Quenching pump.
16: Quench isolating valve (outlet side).
17: Product cooling water manometer (quench side).
18: Product cooling water flow detector (quench side).
19: Quench circuit.
20: Product cooling water tank.

4. Technical features

4.1. Induction hardening machine

4.1.1. Hardening solid state generator

Model	400SM20
Output power	400 kW
Nominal frequency range	0.5-20 kHz
Input voltage	3x480V – 60 Hz
Technology	IGBT Series digital
MF / HF voltage	450V
Energetic efficiency	92%

4.1.2. Oscillating circuit capacitors bank

Working frequency (Bells hardening)	10 / 30 kHz
Working frequency (Stem hardening)	8 / 10 kHz
Working frequency (Female hardening)	~30 kHz

4.1.3. Coils interface transformer/s

Model (Bells hardening)	T4000/30
Model (Stem / Female hardening)	T4000/30

4.1.4. Coil / inductor

Type (Bells / Stem hardening)	Single shot hardening
Includes product cooling quench	YES
Water fastening type	Fast connector
MF/HF electric connection method	Quick mechanical connection (Clamps)

4.1.5. Induction system coolant

Model	R3
Industrial water required temperature	22º C
Industrial water required pressure	2…7 bar
Heat exchanger type	Welded
Additional equipment	Overpressure pump; Stainless steel piping

4.1.6. Product coolant

Model	R4
Industrial water required temperature	22º C
Industrial water required pressure	2…7 bar
Heat exchanger type	Welded
Additional equipment	Paper filter; Stainless steel piping

4.1.7. Product handling system

Operation or stations: ....Hardening

Model or type MBA-321

Number of stations .... 1 Bell hardening + 1 Stem hardening

Functioning principle ....Indexing table 4 stations type

Electric supply: ....3 x 480 V – 60 Hz

Pneumatic supply ....5 ÷ 6 bar

4.1.8. Hardening machine control system

Computer Numeric Control (CNC): ....Siemens 840 D, PCU 50, 9 axles, 3 controlled by CNC

Man - Machine Interface (MMI) ....Siemens OP12

Human held unit (HMI) .... Siemens HT2

4.2. Painting

According to GH Standard, as follows:

Electrical cabinet (inside/outside) and raceways..Grey RAL 7035

Electrical panel .... Zinc coated

Mechanical tooling (base and frame) ....Grey RAL 7035

Pipes .... Green RAL 6018

Machine inner tank ....Moist Metal Grip

Type of painting ....PU Bi-component

Commercial and standard components ....as per suppliers

Other colors or standards requirements may suppose additional charges on project.

5. Commercial offer

5.1. Delivery and guarantee terms

The present offer is in accordance with GH Electrotermia S.A. sales gen. terms & standard GH NG 1.7.

5.2. Acceptation tests

After manufacturing, the equipment here proposed will be tested. The tests will be performed in the GH Electrotermia S.A. factory in San Antonio de Benagéber (Valencia, Spain). GH will confirm to the customer the date of equipment availability for tests. The tests will consist on:

1. Static tests (supply verification)
   1.1. Concordance of manufacturing to the contract .... YES
   1.2. Accordance with safety and ergonomics .... YES
   1.3. Technical documents consistency .... YES
2. Dynamic tests (running verification)
   2.1. Inverter and oscillating circuit .... YES
   2.2. Induction equipment cooling system .... YES
   2.3. Product cooling system .... YES
   2.4. Handling system .... YES
   2.5. Control and command system .... YES
   2.6. Quality process supervision system .... YES
3. Treatment or process verification
   3.1. Treatment or process accordance with specifications .... YES
4. Production
   4.1. Cadence / speed of production .... YES
   4.2. Starting time .... YES
   4.3. Stability of running (H hours test) .... NO
   4.4. Change of product / reference time .... NO

A representative of the buyer must attend the tests together with GH personnel in order to sign the acceptance document. The client may give GH notice of renouncement and the tests will be undertaken without the presence of the client's representative. The reception documents will then be valid as an acceptance document.

This report authorizes the delivery, signed by a customer's representative and a GH's representative, will be stated with the declaration of conformity –and subsequently, the authorization for delivery- or the modifications

5.3. Guarantee

The guarantee period is for 1 year after FAC (however, for 2 years after FAC for parts related to NC - submission of warranty letter) and the supplier undertakes to repair or replace as quickly as possible any parts of the supplies which, before the expiry of the guarantee period are proved to be defective due to bad material, faulty design or poor workmanship. The supplier shall bear the costs of remedying the defective parts in its works. If the repair cannot be carried out in supplier's works, the customer shall bear the related costs to the extent exceeding the customary costs of transport, personnel, travelling, living, dismantling and reassembly of the defective parts.

If dispatch or taking over or erection is delayed due to reasons beyond supplier's control, the guarantee period shall end not later than 18 months after supplier's notification that the supplies are ready for dispatch.

The guarantee expires prematurely if the customer or a third party undertakes inappropriate modifications or repairs or if the customer, in case of a defect, does not immediately take appropriate steps to mitigate the damage and informs the supplier of such defect.

5.4. Delivery terms following Incoterms 2010

FOB Port: Valencia (Spain)

5.5. Payment terms

20 %	Payable upper presentation of A/Bond (by T/T without delay).
65%	Payable upon shipping documents (by irrevocable and confirmed L/C, opened in a first class opened at last 3 months before FOB estimated date).
15 %	Payable upon FAC issued by Hyundai-WIA and Warranty bond (by T/T without delay).

5.6. Delivery time

15th September 2017 FOB Valencia (with a grace period of 15 days).

A Hyundai-WIA parts delay for the trials in GH will correspond in a delay of the delivery time by the same period of time.

5.7. Validity of the proposal

The present proposal is valid for 7 days.

5.8. Prices 7002901.6

(Prices include not taxes). Currency: EURO

Item	Designation
1	Induction hardening equipment, including:
1.1	One (1) induction hardening power supply 400KW, series type. Including cooling by water/air
1.2	One (1) Capacitor bank for bells hardening
1.3	One (1) Capacitor bank for stems hardening
1.4	Two (2) T4000/30 matching transformer
1.5	Two (2) output plates
1.6	One (1) pneumatic switch for stem and bells hardening stations
2	Heat Treat handling machine
2.1	One (1) hardening rotating table concept machine PBA-525 type as described on quotation.
2.2	One (1) set of safety covers and protections
2.3	One (1) common platform (skid format)
3	Control & Command system
3.1	One (1) Control & command system based upon CNC Siemens 840d SL with control panel (OP12) and process parameters control and supervision and HT2 hand held unit.
3.2	One (1) PMS graphic parameter recorder
3.3	One (1) cooling exchanger water/air for electric components.
3.4	One (1) Data & statistics analysis & recording system (Data logger)
3.5	One (1) Teleservice system for remote assistance by means of VPS
4	Pre-acceptance tests at GH Induction facility
5	Delivery and Installation
5.1	Dismantling for transport and packaging
5.2	Final acceptance tests on customer's site
6	Documentation
6.1	Three (3) paper set and e-copy of documentation in Spanish and two (2) in English.

Price in EUROS Items 1-6, Incoterm 2010: EXW (Valencia):	~~925.000,-€~~ *(Discount)* 617.175,-€

COOLING

		Units	Unit price	TOTAL
7	Induction components equipment cooling system GH-R3 type with heat exchanger and overpressure pump for inductors and stainless steel piping.	1	29.000,-€	~~29.000,-€~~ 19.324,-€
8	Quench liquid circuit system GH-R4 type with heat exchangers, heating resistances, stainless steel tank and stainless steel piping. It includes control and command system and paper filter.	1	61.000,-€	~~61.000,-€~~ 40.647,-€
9	Magnetic separator on R4 circuit	1	24.000,-€	~~24.000,-€~~ 15.992,-€

ACCESORIES

		Units	Unit price	TOTAL
10	Washing machine station	1	96.500,-€	~~96.500,-€~~ 64.302,-€
11	Input buffer conveyor	1	11.000,-€	~~11.000,-€~~ 7.330,-€
12	Output buffer conveyor	1	11.000,-€	~~11.000,-€~~ 7.330,-€
13	Eddy current and laser and dot marking station	1	325.000,-€	~~325.000,-€~~ 216.561,-€
14	Spare parts set*	1	32.000,-€	~~32.000,-€~~ 11.667,-€*

**Spare parts set supply for the 3 machines would be valued on 35.000€. Another 34.000€ would be let in stock in GH Mexico facilities during all warranty for its use if required. After that period Hyundai WIA can purchase this set for this value.*

INDUCTORS AND TOOLS

15	Inductor set for Bells in Single-Shot	6	16.250,-€	~~97.500,-€~~ 64.968,-€
16	Spare inductor for Bells in Single Shot	6	9.800,-€	~~58.800,-€~~ 39.181,-€
17	Inductor set for Short Stem in Single-Shot	8	10.000,-€	~~80.000,-€~~ 53.307,-€
18	Spare inductor for Short Stem in Single Shot	8	5.700,-€	~~45.600,-€~~ 30.385,-€
19	Inductor set for Long Stem in Single-Shot	5	18.000,-€	~~90.000,-€~~ 59.971,-€
20	Spare inductor for Long Stem in Single-Shot	5	6.700,-€	~~**33.500,-€**~~ **22.323,-€**
21	Inductor and quench set for Female parts in Scanning	1	8.900,-€	~~8.900,-€~~ 5.930,-€
22	Spare inductor for Female parts in Scanning	1	4.000,-€	~~4.000,-€~~ 2.665,-€
23	Tooling set for coil's check (valid for all references)	1	6.000,-€	~~6.000,-€~~ 3.998,-€

24	Jigs set per reference	6	2.800,-€	~~16.800,-€~~ 11.195,-€

SERVICES

25	Sea-worthy packing and freight FOB Valencia	1	~~20.000,- €~~ **22.988,-€**
26	Start up, installation and training of staff Hardening machine	4 Weeks	~~36.500,- €~~ 24.322,-€
27	Service offer pack (Training pack + 1 month production assistance)	4 weeks	~~32.000,-€~~ * 0,-€

** Service offer will be offered free of cost in case of purchase of the three projects for Mexico*

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TOTAL Price Items 1-27 Incoterm 2010:FOB Valencia (Spain)	1.341.562,-€